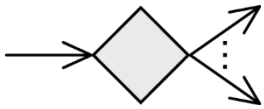
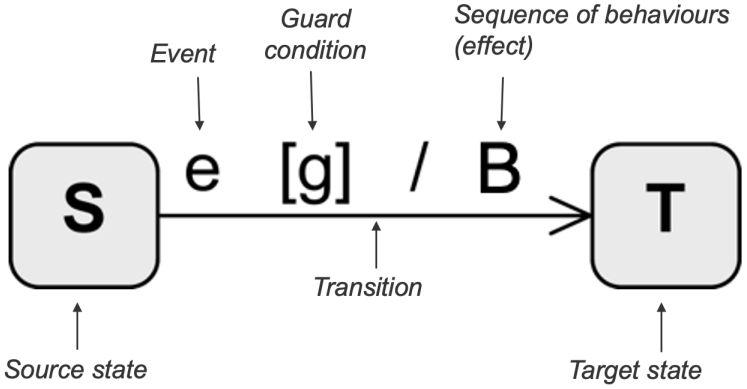
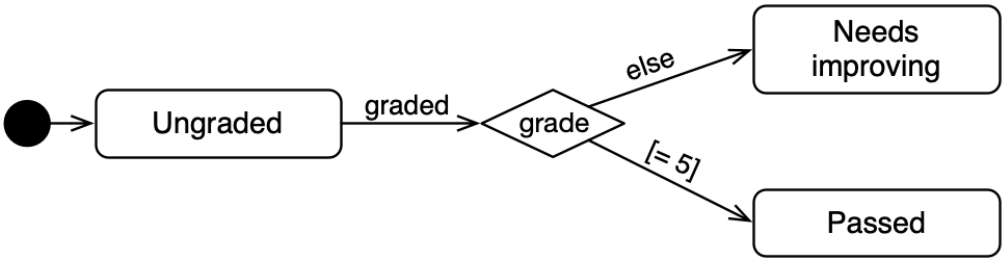
Appendix: State Machine Diagram reference

Notation	Description
S entry/Activity(...) do/Activity(...) exit/Activity(...)	State: Description of a specific “time span” in which an object finds itself during its “life cycle”. Within a state, activities can be executed by the object.
S —e→ T	Transition: Transition **e** from source state **S** to target state **T**
●	Initial State
◉	Final State (remains in this state)
×	Terminate state (object is deleted)
	Decision Node
escapeButtonPressed	Signal event
register(exam)	Call event
after(10 seconds) when(time == 10:00)	Time events
when(temperature < target)	Change event





Some common errors in State Machine submissions:

- Common minor errors:
  - wrong type of arrow-head
  - adding an event for the transition from the initial state (this arrow should not be annotated, apart from "new" if diagram starts when a new object is created)

- Common medium errors:
  - no arrow heads for transition lines
  - having more than one initial state
  - having no initial state
  - having guard conditions but no event for transitions leaving states (generally, the only time you just have a guard condition is for transition from a decision diamond)
  - not having mutually exclusive conditions for the transitions leading out of the decision diamond
  - having the same state appear more than once in the diagram
  - having 'orphan' states, from which there are no transitions (there should not have been in this test)

- Common major errors:
  - no example of decision diamond
  - not having any event for the transition leading into the decision diamond